Capstone Project Write-Up

Priced Out: Predicting the Next U.S. Housing Crisis

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1 Title, Team, and Affiliations

Project title: *Priced Out: Predicting the Next U.S. Housing Crisis*

Name	Role	Willamette email
Brandon Smith	Owner Product Lead	bwsmith2@willamette.edu

Peer Stakeholder POs: *Manish Reddy Kallu / Alexander Jackson / Jackson Garro*

Repository: `https://github.com/BSmith-cpu/DATA-510-CAPSTONE-`

2 Abstract

(150 to 250 words after you have results. Summarize: problem, approach, key finding, impact, limitation. Write this section last.)

3 Introduction: Problem and Research Questions

Problem statement:

- Many U.S. metro cities are dealing with the issue of forcing out the middle class. This is a combination of many factors (house values, median income, etc). We are looking at cities already affected by this (Boise, Austin, Tampa) and will use them as training cities to see whether we can identify predictors from that vast data source.

Primary research question:

- Which combination of housing affordability and demographic indicators best predict declines in the share of middle-income households in U.S. metro areas, and how do these predictors appear in Boise City, Idaho, Austin, Texas, and Tampa, Florida between 1975 and 2025 (50 year time span).

Secondary questions (if any):

- TBD

Changes since M1 proposal: *(none yet, or explain pivot)*

- None (06/08/2026)
-

4 Impact and Stakeholders

5 Related Work and Background

Wildfire smoke and respiratory health are well studied at regional scale (Reid et al. 2016; Liu et al. 2016). This project applies county-level public data to Oregon specifically.

6 Data and Data Engineering

Sources:

Source	Role	Access

Engineering summary: Raw data in `data/raw/`; processed panel in `data/processed/`; ingest scripts in `src/ingest/`.

```r
# Replace with your project data when ready:
# panel <- readRDS(here::here("data/processed/analysis_panel.rds"))
# nrow(panel)
knitr::kable(data.frame(
  check = c("Rows", "Counties", "Date range"),
  status = c("stub", "stub", "stub")
))
```

Table 3: Example QC summary (replace with your panel)

check	status
Rows	stub
Counties	stub
Date range	stub

7 Ethics and Responsible Use

- **Privacy:**
- **Fairness / equity:**
- **Limitations of setting:**
- **Non-causal framing:** *(if observational)*

8 Methods and Evaluation

Design:

Primary analysis:

Evaluation metrics:

Train / validation / holdout:

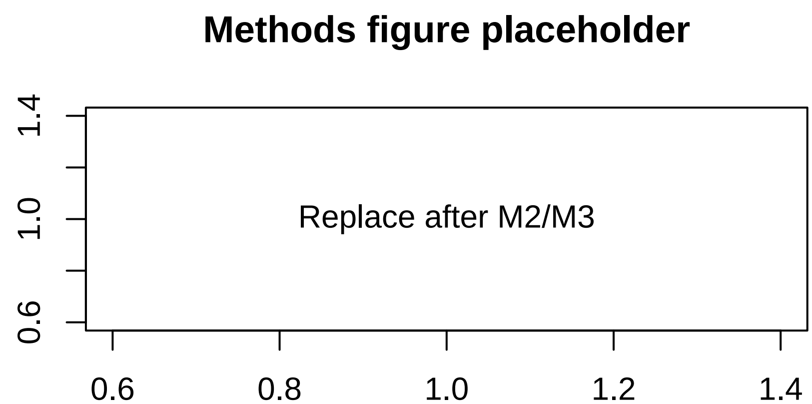

Figure 1: Replace with a methods diagram or exploratory plot

9 Results and Visualizations

(Stub: key findings with Figure 1 and tables.)

10 Discussion, Limitations, and Future Work

Interpretation:

Limitations: *(tie to actual data and design)*

Future work:

11 Conclusions and Recommendations

(Short, stakeholder-facing takeaways.)

12 Reproducibility and References

Reproduce main results:

```
cd your-repo
quarto render deliverables/M4-writeup-draft/writeup.qmd
```

Dependencies: `requirements.txt` or `renv.lock` in repo root.

Data: URLs and approval status documented in `docs/data-dictionary.md`.

13 References

Liu, J. C., L. J. Mickley, M. P. Sulprizio, et al. 2016. "Wildfire-Specific Fine Particulate Matter and Risk of Hospital Admissions in Urban and Rural Counties." *Epidemiology* 27 (3): 350–57.

Reid, C. E., M. Brauer, F. H. Johnston, et al. 2016. "Critical Review of Health Impacts of Wildfire Smoke Exposure." *Environmental Health Perspectives* 124 (8): 1334–43.